General information

Title : LED driver
Eng : Michael
Master description revision : v2 (09/14/2025)

Change sheet

When?	What?
12/02/2025	Created the document, filled in basic info. Concept description and initial analysis.
12/02/2025	Define HLR for HW & FW, define LLR for FW, define block schematic.
01/05/2026	Finalize hardware LLR.
01/06/2026	Oscillator design.

References

Master description file
https://github.com/DevxMike/kitchen_driver/blob/main/Master_description_file_v2.pdf

AN2867
https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf

Brief

This document describes the flexible low-level driver part of the project. As per the frozen config (see the references) this block should be implemented as an independent part, independent embedded system with basic functionality and communication port. This section describes high-level requirements towards hardware and firmware sourcing from system requirements in master config file.

Requirements towards hardware

HLR_ID	SYS_ID	Description
HLR_HW_1	SYS_2 SYS_5	Board must utilize UART or USB port for communication purposes.
HLR_HW_2	SYS_3	As per details in the master config, the board must handle power regulation and distribution.
HLR_HW_3		Board must be able to power an SBC, LED and logic.
HLR_HW_4	SYS_4	Board must be able to exchange data with LED strip (WS2811 driver).
HLR_HW_5		Board must introduce elements enabling user to interact with.
HLR_HW_6	SYS_1	Microcontroller should be clocked with a stable timebase source.
HLR_HW_7		Firmware should boot from flash.
HLR_HW_8		Design should allow to easily attach programmer / debugger to the microcontroller.

NOTE: Because this project is a separate independent embedded system, it is assumed, that **SYS_1** requirement is satisfied.

Requirements towards firmware

HLR_ID	SYS_ID	Description
HLR_FW_1	SYS_2 SYS_5	Firmware must use the same packet description as the server side. It must come with methods for serializing and deserializing packets to act accordingly.
HLR_FW_2		Firmware must handle handshake and link status periodically.
HLR_FW_3		Firmware must handle packets containing settings from the server device.

HLR_FW_4	SYS_4	Firmware must handle communication with LED strip (WS2811 driver)
HLR_FW_5		Firmware must handle basic animations and functionalities (ON, OFF, BRIGHTNESS / COLOUR SETTING).
HLR_FW_6		Firmware must handle user interface.

Detailed hardware description

As per the master description, the board should incorporate STM32 microcontroller. For that, STM32L152 series microcontroller will be used as it introduces more than enough:

- 80kB RAM
- 11 timers
- USB and USART ports.

This part comes in both standalone chip and on nucleo development board hence makes the solution easy to develop, debug and port prototype to a board.

Low-level requirements

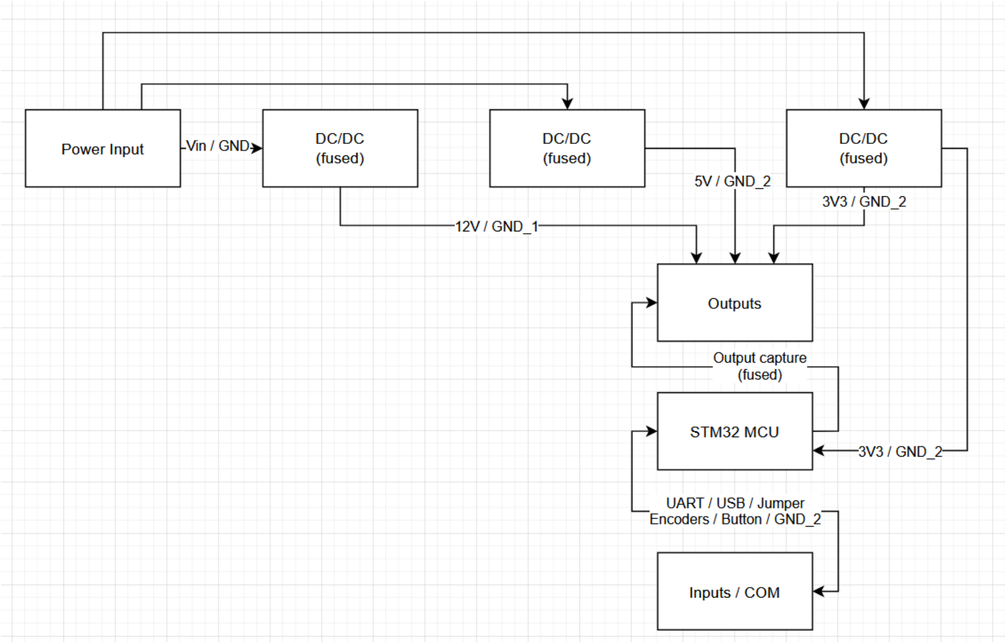
LLR_ID	HLR_ID	Description
LLR_HW_1	HLR_HW1	Board must introduce a jumper to select active port (USB; **PA12 and PA11** or UART2; **PA2 and PA3**) by switching **PB11** between GND and Vcc.
LLR_HW_2		If **PB11** pin is shorted to GND, UART is enabled as active port.
LLR_HW_3		If **PB11** pin is shorted to Vcc, USB is enabled as active port.
LLR_HW_4		UART should be configured as follows: 115200 baud, 8 data bits, 1 stop bit, parity none.
LLR_HW_5	HLR_HW_2 HLR_HW_3	Board must introduce impulse DC/DC converter to convert input voltage to 12V.
LLR_HW_6		12V line should be available in the output bar (supply for LED strip and auxiliary supply line).
LLR_HW_7		12V line should be minimum 2cm wide.
LLR_HW_8		Board must introduce impulse DC/DC converter to convert input voltage to 5V.
LLR_HW_9		5V line should be available in the output bar.
LLR_HW_10		5V line should be minimum 1cm wide.
LLR_HW_11		Board must introduce impulse DC/DC converter to convert input voltage to 3V3.
LLR_HW_12		3V3 line should be available in the output bar.
LLR_HW_13		3V3 line should be minimum 0.5cm wide.
LLR_HW_14		12V LED supply line should be fused up to 16A.
LLR_HW_15		12V auxiliary supply line should be fused up to 5A
LLR_HW_16		5V auxiliary supply line should be fused up to 5A.
LLR_HW_17		3V3 logic supply line should be fused up to 0.1A.
LLR_HW_18		3V3 auxiliary supply line should be fused up to 1A.
LLR_HW_19		Board must introduce a switch that allows physical power cutoff.
LLR_HW_20		3V3 logic supply (3V3 and GND) must be separated 12V supply lines.

LLR_ID	HLR_ID	Description
LLR_HW_20	HLR_HW_4	Output compare signal should be available in the output sections (TIMER 4, CHANNEL1, PB6).
LLR_HW_21		MCU should drive LEDs via optocoupler to ensure that supplies are separated.
LLR_HW_22	HLR_HW_5	Incremental encoder A should be attached to EXTI Line (**PA8 PA9** encoder; **PA10** encoder button).
LLR_HW_23		Incremental encoder B should be attached to EXTI Line (**PB13 PB14** encoder; **PB15** encoder button).
LLR_HW_24		**PB11** input should be configured as a pull-up and available as a jumper.
LLR_HW_25	HLR_HW_6	HSE pins should be attached to 8MHz cristal oscillator.
LLR_HW_26	HLR_HW_7	**BOOT[1:0]** pins should be attached to **GND** via pull-down resistors.
LLR_HW_27	HLR_HW_8	Board should expose: **Vdd** from target, **SWCLK(PA14)**, **GND**, **SWDIO(PA13)**, **NRST**, **SWO (PB3)**.

Testing methodology

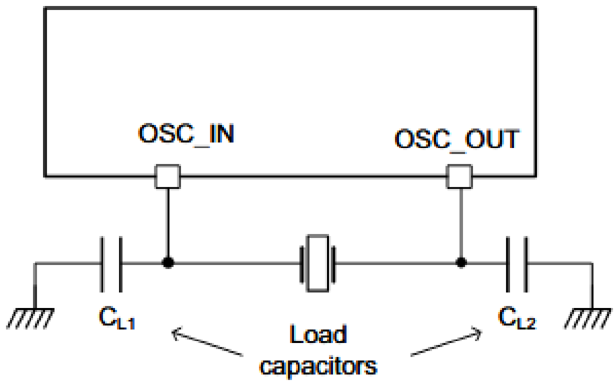
a) electrical tests - shorts, supply availability
b) functional tests - output compare oscilloscope test, encoder test, jumper test, USB and UART tests

Block schematic of electronic board


Power Input
Vin / GND
DC/DC
(fused)
DC/DC
(fused)
DC/DC
(fused)
5V / GND_2
3V3 / GND_2
12V / GND_1
Outputs
Output capture
(fused)
STM32 MCU
3V3 / GND_2
UART / USB / Jumper
Encoders / Button / GND_2
Inputs / COM

Oscillator design

Crystal/Ceramic resonators



MSv42081V1

Table 6. HSE oscillators embedded in STM32 MCUs/MPUs

Series	F0 F3	F1 T	F2	F4	F7	L0	L1	C0 L4, L4+ L5 H7 G0, G4 MP1 U0	H5 U3 U5	MP2	N6	Unit
Frequency range	4 - 32	4 - 16	4 - 25	4 - 26	4 - 26	1 - 25	1 - 24	4 - 48	4 - 50	16 - 48	16 - 48	MHz
g_m (min)	10	25	5	5	5	3.5	3.5	7.5	7.5	12.5	9.57	mA/V
$G_{m\_crit\_max}$	2	5	1	1	1	0.7	0.7	1.5	1.5	2.5	1.95	

YIC Quartz Crystal Unit (49US/49SL/49SSL/49SMT/49SLMT/49SSLMT)

49US ,49SL ,49SSL Series: HC-49US ,HC-49SL ,HC-49SSL
49SMT ,49SLMT ,49SSLMT Series: HC-49SMT ,HC-49SLMT ,HC-49SSLMT SMD Type

3.000 - 80.000MHz

● SPECIFICATION

Frequency Range	3.000 - 80.000MHz
Vibration Mode	AT-Cut fundamental , third , fifth , seventh
Frequency Tolerance (25℃)	±10,±20,±30PPM
Operation Temperature	-20℃ ~ +70℃
Shunt Capacitance	7pF max or Special
Load Capacitance	Series ,16 , 20 , 22 , 30pF or Special
Drive Level	50,100,300,500,1000 μW
Insulation Resistance	500MΩ min/DC
Aging	±3,±5PPM/year

● EQUIVALENT SERIES RESISTANCE

Frequency (MHz)	ESR(Ω)max
Fundamental	
3.000 - 3.799MHz	180
3.800 - 4.999MHz	150
5.000 - 5.999MHz	120
6.000 - 7.999MHz	80
8.000 - 9.999MHz	80
10.000 - 11.999MHz	50
12.000 - 15.999MHz	50
16.000 - 19.999MHz	40
20.000 - 35.000MHz	40
3rd Overtone	
26.000 - 39.999MHz	100
40.000 - 80.000MHz	80

https://www.tme.eu/pl/details/8.00m-smdhc49s/rezonatory-kwarcowe-smd/yic/

$$C_L = \frac{C_{L1} \times C_{L2}}{C_{L1} + C_{L2}} + C_S$$

$$C_{L1} = C_{L2}$$

$$C_L = \frac{C_{L1}^{2}}{2 \cdot C_{L1}} + C_s$$

$$C_L - C_S = \frac{C_{L1}^{2}}{2 \cdot C_{L1}}$$

$$20pF - 7pF = \frac{C_{L1}^{2}}{2 \cdot C_{L1}}$$

$$13pF = \frac{C_{L1}}{2}$$

$$26pF = C_{L1}$$

Detailed firmware description

Low-level requirements

NOTE: It has been decided, that communication between threads and task scheduling will be handled by freeRTOS framework.

LLR_ID	HLR_ID	Description